
MP Online • Internship Project • 2025-26

GrievanceHub

Digital Grievance Redressal System

Development Planning Document

Project Schedule, Tools & Team Roles

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MP Online Internship Project

1. Project Overview

GrievanceHub is a Digital Grievance Redressal System developed as part of the MP Online Internship Program. The system enables citizens to submit complaints regarding public services and tracks them through to resolution.

The project was developed using a modern full-stack architecture: FastAPI (Python) on the backend, React with Tailwind CSS on the frontend, and MySQL as the database.

2. Development Timeline

Phase	Tasks	Duration	Status
Phase 1 - Planning	Requirement gathering, BRD, system design	Week 1	Completed
Phase 2 - Backend Setup	FastAPI project init, DB models, auth system	Week 2	Completed
Phase 3 - Core APIs	Grievance submission, ticket generation, admin APIs	Week 3	Completed
Phase 4 - Frontend	React pages, routing, Tailwind UI, Axios integration	Week 4	Completed
Phase 5 - Testing	Unit testing, API testing via Swagger, bug fixes	Week 5	Completed
Phase 6 - Review	Documentation, code cleanup, final submission	Week 6	Completed

3. Technology Stack

Layer	Technology	Version	Purpose
Backend Framework	FastAPI	0.115.0	REST API development
Server	Uvicorn	0.30.1	ASGI server for FastAPI
Database ORM	SQLAlchemy	2.0.30	Database abstraction layer
Database	MySQL	8.0+	Persistent data storage
Auth Library	python-jose	3.3.0	JWT token generation & validation
Password Hashing	Passlib + bcrypt	1.7.4 / 4.0.1	Secure password storage
Data Validation	Pydantic v2	2.10.6	Schema validation & serialization
Frontend Framework	React	18+	UI component development
Build Tool	Vite	Latest	Frontend bundler & dev server
Styling	Tailwind CSS	3.x	Utility-first CSS framework
HTTP Client	Axios	Latest	API calls from frontend
DB Driver	PyMySQL	1.1.1	MySQL connector for Python

4. Team Roles & Responsibilities

Role	Assigned To	Responsibilities
Full Stack Developer	Shantanu	End-to-end development of backend APIs and frontend UI
Backend Developer	Shantanu	FastAPI setup, DB models, JWT auth, all routers and services
Frontend Developer	Shantanu	React pages, routing, Axios integration, UI components
Database Designer	Shantanu	MySQL schema design, SQLAlchemy model definition
Tester	Shantanu	Swagger UI testing, manual test cases, bug verification
Documentation Lead	Shantanu	BRD, DDD, RTM, Test Docs preparation

5. Development Tools

Tool	Category	Usage
VS Code	IDE	Primary code editor
Postman / Swagger UI	API Testing	Testing all REST endpoints
MySQL Workbench	Database	Schema management and query testing
Git + GitHub	Version Control	Source code management
npm / pip	Package Manager	Dependency management
Vite Dev Server	Frontend Dev	Hot reload during UI development